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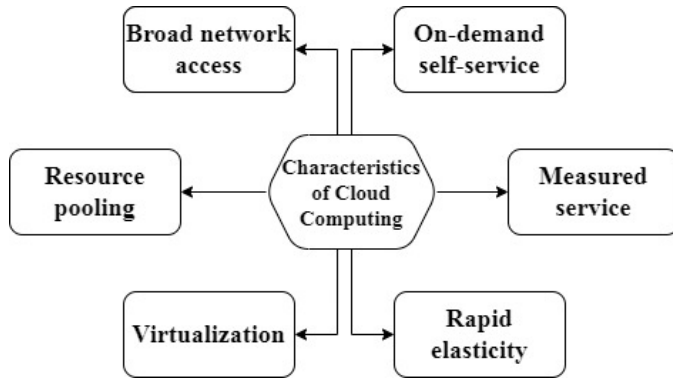


Fig. 2. Characteristics of Cloud Computing.

the master-master replication, multiple nodes act as master nodes and can receive write requests. This can improve write performance and availability, but can also introduce data consistency issues [5]. In multi-master replication, all nodes are capable of receiving and replicating write requests, which can improve write performance and availability, but requires more complex synchronization mechanisms to maintain data consistency. Overall, replication of data is an essential technique for ensuring data availability, reliability, and performance in distributed cloud environments. It is significant to choose the appropriate replication strategy based on the specific requirements and constraints of the application and environment. When designing a replication of data strategy for a distributed cloud environment, it also is essential to consider factors such as data consistency, latency, and bandwidth requirements. Choosing the right replication method and configuring it correctly can help ensure that customer data remains available and accessible to users at all times. In this article, we will survey a few of the dynamic replication of data methods employed in the cloud environment. Many studies have been conducted on these two categories of replication of data. In order to enhance the effectiveness, performance, and dependability of dynamic replication of data in cloud, the main focus will be on coming up with a better solution than the previous findings.

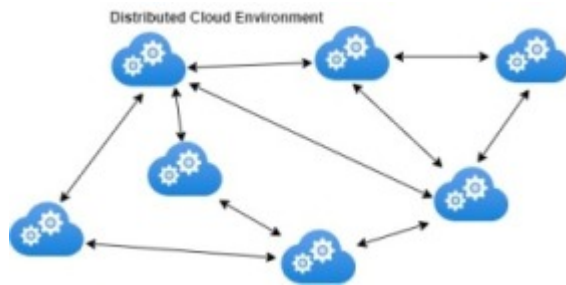


Fig. 3. Distributed Cloud Environment.

A. Dynamic Replication of Data

Dynamic replication of data in cloud refers to the task of automatically producing and maintaining several copies

of same data in different locations to guarantee reliability, availability and performance. Figure 4 represent the process of replication of data. Cloud computing platforms, such as Microsoft Azure, and Google Cloud Platform (GCP), Amazon Web Services (AWS) offer various options for replication of data, including synchronous and asynchronous replication. Intelligent decisions concerning the location of data made by dynamic replication of data are based on knowledge of the present environment [6]. Figure

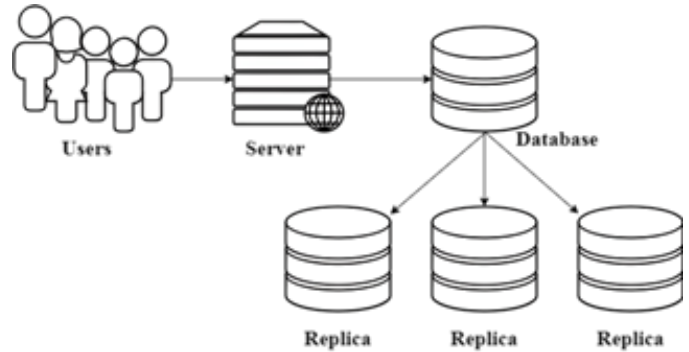


Fig. 4. Replication of data.

Figure 3 represents the replication of data. A service-specific environment, where the quantity and location of users who will access data must be determined in a very dynamic way, is where a dynamic replication technique is most suitable [7]. By offering a variable number of copies in the data cloud system, it may optimize both efficiency and resource utilization. I/O and communication costs are combined to optimize total cost. Dynamic replication is significantly higher and more efficient than static replication due to the process of intelligent higher cognitive and determining the placement of the duplicate according to the demand by taking into account the surrounding condition. Replication of data is the process of creating and maintaining one or more copies of data [8]. There are several types of replication of data, including:

- Full or complete replication: In full or complete replication, every data are replicated to each replica or copy. This ensures that every replica have the same data as the original copy. However, it can be resource-intensive and can consume a significant amount of storage space [9].
- Partial replication: In partial replication, simply a subset of the data is replicated to each replica. This is often used when certain data is more critical than others, and it is not necessary to replicate all data.
- Snapshot replication: In snapshot replication, a snapshot of the data is taken at a specific point in time and then replicated to other locations. This type of replication is useful for data that changes infrequently or when the data is too large to replicate regularly.
- Transactional replication: In transactional replication, changes made to the original data source are replicated in real-time to other locations. This type of replication is useful when changes to data need to be propagated

quickly.

- Merge replication: In merge replication, changes made to the original data source and replicas are merged together periodically. This type of replication is useful when multiple copies of data are updated frequently, and changes can be made at any location.
- P2P replication: In P2P replication, data is replicated between multiple nodes without a central server. This type of replication is useful when there are multiple replicas, and changes need to be propagated between them quickly.
- Master-Slave Replication: In master-slave replication, there is one primary (master) database that is responsible for processing all write operations, while one or more secondary (slave) databases replicate data from the master. The primary database acts as the source of truth, and the secondary databases are used for read-only operations, backups, and disaster recovery.
- Multi-Master Replication: In multi-master replication, multiple databases act as the primary source for write operations, and changes made in any of the databases are automatically replicated to all the other databases. This approach provides high availability and allows for scaling write operations, but it can be more challenging to maintain consistency and resolve conflicts.
- Cascading Replication: Cascading replication involves chaining multiple replication processes together, where data is first replicated from one database to another, and then from that database to another, and so on. This approach can be used to replicate data across multiple geographically dispersed locations, but it can also introduce latency and increase the likelihood of data inconsistency. The type of replication of data used depends on the specific use case and requirements of the organization [10].

B. Replication of data Tools and Utilities

There are many replication of data tools available for use in the cloud, and new tools are constantly being developed [11]. Here is a non-exhaustive list of some of the most popular and widely used replication of data tools in the cloud.

- Apache Kafka: Kafka is a one kind of distributed streaming platform that can be used for real-time replication of data. It can handle large volumes of data and provide high availability and fault tolerance.
- Amazon S3 Replication: This is a feature provided by Amazon Web Services (AWS) that allows users to replicate data across different S3 buckets in different regions. It provides both cross-region and same-region replication.
- Google Cloud Storage: Google Cloud Storage provides multi-regional and regional replication options. Users can choose to replicate data across multiple regions or within a single region.
- Microsoft Azure Storage: Azure Storage provides different types of replication options including locally redundant storage (LRS), zone-redundant storage (ZRS), geo-redundant storage (GRS), and geo-zone-redundant

storage (GZRS). These options provide different levels of replication and redundancy for data.

- Apache Hadoop: Apache Hadoop is a free and open-source framework that provides distributed storage and processing of large datasets. It has a built-in replication of data mechanism that can replicate data across different nodes in a Hadoop cluster.
- MongoDB: MongoDB is one of the most popularly known NoSQL database that provides built-in replication and sharding capabilities. Users can replicate data across multiple nodes in a MongoDB cluster to ensure high availability and fault tolerance.
- Cassandra: Apache Cassandra is another popular NoSQL database that provides built-in replication capabilities. Users can replicate data across multiple nodes in a Cassandra cluster to ensure high availability and fault tolerance.
- GlusterFS: GlusterFS is an open-source distributed file system that provides replication and distributed storage capabilities. It can be used to perform replication of data among the different nodes in a distributed cloud environment.
- Azure Blob Storage Replication: This feature of Azure Blob Storage allows you to replicate data across different regions, storage accounts or data centers. It provides options to configure replication policies, including geo-redundant and read-access geo-redundant storage.
- Rsync: This is a widely used utility for synchronizing files and directories between different machines. It can be used to replicate data across cloud environments, although it requires manual configuration and monitoring.
- Apache Storm: Storm is a another one kind of distributed real-time computation system that can be used for replication of data. It provides a fault-tolerant and scalable platform for processing data streams and replicating them across multiple clusters.
- Apache Hadoop : Hadoop's NameNode manages the file system namespace and tracks the location of each data block in the cluster, while the DataNodes store the actual data blocks. Hadoop replicates data blocks across multiple DataNodes for fault tolerance, and the number of replicas can be configured based on the specific needs of the organization.
- Oracle Data Guard: It can be used for replication of data in a distributed environment. Data Guard provides a comprehensive high-availability and disaster recovery solution for Oracle databases, including the ability to replicate data between primary and standby databases. Data Guard uses real-time redo apply to replicate data changes from the primary database to one or more standby databases.
- Zerto Virtual Replication: It is a replication of data and disaster recovery solution that is designed to provide continuous availability and data protection for virtualized and cloud environments. Zerto Virtual Replication uses continuous data protection (CDP) technology to replicate

data in near real-time between primary and secondary sites, enabling rapid recovery in the occasion of a disaster or outage.

These tools can be used depending on the specific requirements of the distributed cloud environment and the type of replication of data needed. The choice of tool depends on factors such as the specific use case, the size and complexity of the data, and the specific requirements for replication such as latency, consistency, and durability [12].

II. RELATED WORKS

Figure 5 represents the process involve in replication of data in distributed cloud environment. some of the work related to the data replication is summarized in the table 1. From the survey, it is found that the algorithm must ensure that the data is replicated across multiple nodes in a distributed cloud environment, ensuring data availability and reliability. By replicating data across multiple nodes, the system can continue to function even if one or more nodes fail, which reduces the risk of data loss or downtime. The algorithm should also ensure data consistency by synchronizing the replicas periodically, which ensures that all replicas contain the same data.

III. PERFORMANCE METRICS

In cloud replication of data strategy, a performance measure is a metric used to assess the efficiency and effectiveness of the replication process. The primary goal of replication of data is to ensure data availability, durability, and consistency [17]. Therefore, performance measures should be designed to evaluate how well the replication strategy achieves these objectives [18]. Here are some of the performance measures that can be used in cloud replication of data strategy:

- Replication time: This metric measure the time duration it requires for data to be replicated from the source system to the target system. A shorter replication time means that data is being replicated quickly, which is important for ensuring that data is up-to-date and available for use.
- Replication frequency: This metric measure how often data is replicated from the source system to the target system. A higher replication frequency means that data is being updated more frequently, which is important for ensuring that data is accurate and up-to-date.
- Data transfer rate: This metric measures the rate at which data need to be transferred from the source system to the target system. A higher data transfer rate means that data is being transferred more quickly, which is important for ensuring that data is available for use in a timely manner.
- Data integrity: This metric measures the accuracy and completeness of the data that is being replicated. It is important to ensure that data is being replicated accurately to avoid data corruption or loss.
- System availability: This metric measures the amount of time that the replication system is available for use. A higher system availability means that the system is more

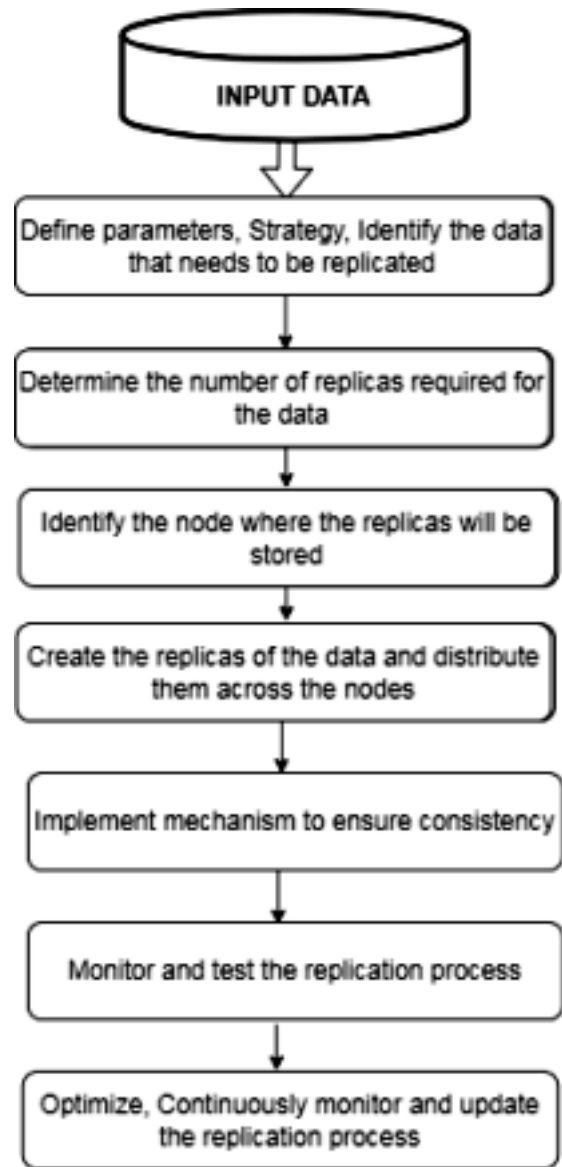


Fig. 5. Working flow of the replication of data in distributed cloud environment.

reliable and can be relied upon to ensure that data is available when it is needed.

- Recovery time objective (RTO): This metric measures the amount of time it takes to recover from a failure or outage in the replication system. A shorter RTO means that the system can be brought back online more quickly, which is important for minimizing downtime and ensuring that data is available for use.
- Recovery point objective (RPO): This metric is used to measure the quantity of data that can be lost in the event of a failure or outage in the replication system. A lower RPO means that less data will be lost in the event of a failure, which is important for ensuring that data is available and up-to-date.
- Replication latency: This refers to the time taken for

TABLE I
RELATED WORK

Author	Proposed Work	Description
Motaz et. al.	Dynamic Replication strategy using Machine Learning Clustering (DRPMLC) on HDFS	Employs machine learning to group files into different groups and use different replication policies to every group to minimise storage utilization, speed up input and output operations, and maintain the reliability and availability of HDFS as a High-Performance Distributed Computing system (HPDC) [13].
Rashed Salem et al.	MOABC algorithm	The various costs and shortest routes in the Cloud in this work using Multi-Objective Optimization (MOO), and they assess the cost distance by utilizing the knapsack problem. The knapsack strategy has been utilised to overcome these challenges in accordance with the distance or quickest routes and lower expenses. [14].
Mansouri et al.	multiple-objective optimized placement algorithm with support of a fuzzy system and meta-heuristic technique	Balances the trade-offs between the six optimization objectives to determine the best places for replicas such as latency, service time, load, system availability, energy consumption and centrality. The number of replicas is calculated without significantly affecting performance to address this issue [15].
Ebadi et al.	hybrid metaheuristic technique to resolve replication problem	To find high-quality solutions, this approach combines the local search capability of the Tabu Search (TS) algorithm with the global search capability of the PSO (Particle Swarm Optimization) algorithm [16].
S. Gopinath et al.	weighted dynamic replication of data strategy for data storage	Each piece of data is given a weight based on how frequently users access it. Each data item's access count, weight, and current replication factor are used to construct a popularity index for it. The data is then categorised as cold, hot or warm based on its popularity index, weight, and a computed threshold value. Dynamically determined replication factors are used for hot and warm data. [17].
Kumar et al.	SWORD approach	A workload-aware placement of data and replication scheme to reduce consumption of resource in such a distributed environment is proposed. Anticipated workload is tracked, represented as a hypergraph, and create partitioning strategies that reduce the average query span, or the average number of machines used to execute a query or a transaction [18].
Sun et al.	Replication strategies appropriate for parallel distributed computing systems	It entails: 1) examining and simulating the correlation between the quantity of replicas and system availability; 2) assessing and finding the most popular data and initiating a replication process when the popularity data exceeds a dynamic threshold; 3) determining the appropriate number of copies to satisfy a sensible system byte effective rate requirement and evenly distributing replicas among data nodes; and 4) designing the dynamic replication of data technique in a cloud. Outcomes from experiments show the better system made possible by the suggested technique in a cloud to be efficient and effective [19].
Wenhao et al.	Multiple-objective offline strategy for optimization to replica management	With an improved artificial immune algorithm that develops a set of solution candidates through cloning, mutation, and selection processes, it decides on the replication factor and replication layout. By evaluation of the trade-offs between the five optimisation objectives, the suggested algorithm known as Multiple-Objective Optimized Replication Management (MORM) looks for the nearly optimal solutions. This article details several experiments that demonstrate the MORM's efficacy [20].
Li et al.	Cost-effective Incremental Replication (CIR)	A brand-new, cost-effective dynamic replication of data approach for cloud data centers is introduced [21]. It is a one kind of data reliability method for cloud-based applications to manage the problem of data dependability in a data centre cost-effectively. The goal of CIR is to adhere to the criteria for data dependability while using the fewest possible replicas. When the existing number of replicas is no longer sufficient to satisfy the data dependability requirement, a new replica needs to be made, according to an incremental replication process implemented in CIR. The minimum data replica number is initially configured by default to be 1.

changes made to the source data to be replicated to the target data. It is important to minimize replication latency to ensure that data is up-to-date and consistent across all systems.

- Replication throughput: This is a measure of the quantity of data that can be replicated in a given period of time. It is significant to maximize replication throughput to ensure that the replication system can keep up with the rate of change in the source data.
- Reliability: This refers to the ability of the replication system to operate continuously without interruption or failure. A good replication system should be reliable to ensure that data is always available when needed.
- Cost: This refers to the total cost of ownership of the replication system, including hardware, software, and maintenance. A good replication system should provide a good balance between cost and performance.

IV. RESEARCH GAP

A research gap in cloud replication of data strategies is the lack of an effective way to verify the accuracy of replicated data. Currently, cloud replication of data strategies rely on the replication of data from one cloud platform to another and trust that the replicated data is accurate. However, there is no reliable way to verify the accuracy of the replicated data, resulting in a potential risk of data loss or corruption [19]. Additionally, cloud replication of data strategies typically lack the ability to detect and resolve conflicts between the source and destination data. As a result, organizations may not be aware of discrepancies between the source and destination data until it is too late. Another one potential research gap in cloud replication of data strategy is the development of a more comprehensive and effective approach to managing conflicts that arise during replication. Conflicts can occur when data is updated in both the source and target systems simultaneously, resulting in inconsistencies that can be difficult to resolve. Current approaches to conflict resolution in cloud replication of data typically rely on simple rules, such as "last write wins" or "highest priority wins," which may not always produce the desired outcome. These approaches can result in data inconsistencies, lost data, or other issues that can impact the reliability and accuracy of the system [19]. Future research could focus on the development of more sophisticated conflict resolution strategies, such as multi-version concurrency control (MVCC) or optimistic concurrency control (OCC). These approaches can allow multiple updates to occur simultaneously while maintaining data consistency, and may be better suited to the dynamic and distributed nature of cloud environments [20]. Another potential research gap is the development of more efficient and scalable replication strategies for large-scale data sets. Replicating large amounts of data can be time-consuming and resource-intensive, which can impact system performance and scalability. Future research could focus on developing more efficient replication algorithms that can handle large data sets more effectively, such as incremental replication or data compression techniques. Finally, research could also

explore the use of machine learning and artificial intelligence techniques to optimize cloud replication of data strategies [21]. These techniques can be used to analyze data patterns and usage trends, predict future data needs, and optimize replication strategies accordingly. This approach could potentially improve system performance and reduce the risk of data loss or inconsistency.

V. OPEN PROBLEMS TO BE SOLVED

One of the important open problem in cloud replication systems is to achieve consistency and availability in the presence of network partitions, also known as the CAP theorem. The CAP theorem states that in a distributed system, it is impossible to simultaneously guarantee all of the following three properties: Consistency, Availability, and Partition tolerance. Consistency refers to the requirement that all replicas of a piece of data in a distributed system must be kept in sync. Availability refers to the ability of the system to remain responsive to requests, even in the face of failures. Partition tolerance refers to the ability of the system to continue operating in the face of network partitions. Cloud replication systems typically rely on techniques such as leader election, quorum-based replication, and consensus algorithms to achieve consistency and availability. However, these techniques are often complex and can lead to performance and scalability issues. Therefore, the challenge is to design cloud replication systems that can handle network partitions while maintaining consistency and availability, without sacrificing performance and scalability. This remains an open problem in the field of distributed systems and cloud computing. Other open problems are listed below.

- Finding an efficient and cost effective replication of data strategy for cloud storage: With the increasing popularity of cloud storage, a major problem is to find an efficient and cost effective replication of data strategy that guarantees data availability and reliability in the cloud.
- Developing a secure and reliable replication of data system for cloud computing: As cloud computing technology has evolved, the need for secure, reliable, and scalable replication of data systems has become increasingly important.
- Understanding the impact of replication of data in performance of cloud: Replicating data in the cloud increases the amount of data stored, and can potentially have a significant impact on cloud performance. It is significant to understand the implications of various replication strategies on cloud performance.
- Optimizing the replication of data process in the cloud: Optimizing the replication of data process in the cloud is challenging, due to the complexity of the underlying systems and the need to balance performance, reliability, and cost.
- Developing approaches for replication of data in multi-cloud environments: Multi-cloud environments present unique challenges in terms of replication of data. It is

important to develop strategies that allow for efficient and reliable replication of data in multi-cloud environments.

VI. FUTURE DIRECTIONS

It is evident from this literature review that there is still more work to be done in the area of cloud data storage. In this part, some of the key aspects that should be taken into account when replicating data are covered. An essential component is the replication decision-making process. Centralized or decentralized replication decisions are also possible. In centralized systems, there is a possibility of a bottleneck if the network is under higher than normal load, and in distributed systems, there is a possibility of pointless replications. The availability of data is found to grow and response time is found to decrease for almost all replication procedures. Most replication algorithms do not take into account optimal bandwidth utilization; however, some strategies do. Sometimes the enhanced data accessibility comes at the expense of increasing bandwidth usage. Another crucial aspect to think about when choosing a replication strategy is the amount of storage space it requires. Some of the techniques made sure that less storage was needed by keeping the right amount of replicas. It is discovered that there is no one method that solves every problem associated with replication of data. While some techniques concentrated on maintaining network capacity, others focused on ensuring availability, reliability, fault tolerance, and load balancing. It is necessary to develop a thorough technique that takes into account all the factors required for better replication of data. The majority of the tactics evaluated the algorithms through simulation. To give a meaningful evaluation of the techniques, these systems must be prototyped and tested in real-world circumstances in the future.

VII. CONCLUSION

In a cloud storage system, replication of data across multiple nodes ensures data reliability and high availability. Dynamic replication of data is more effective than static replication of data because it takes changing patterns of data access into account. This study reviews and compares many suggested ways for each strategy. When replicating data, there are a number of factors that must be taken into consideration, including load balancing, bandwidth consumption, fault tolerance, reduced response times, faster data access, and availability, reliability, scalability, and scalability. This review has shown that there isn't a single method that solves every problem associated with replication of data. Therefore, it is imperative to create a replication of data strategy for cloud storage that takes into account all crucial replication of data factors.

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